

CIV102 Project Team 402 Design Report

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Introduction

The purpose of this project is to design and construct a matboard bridge to carry increasing loading cases. This report will outline the design process used to strengthen the initial bridge, design zero, and provide justification for each of the design decisions made.

The final bridge design is a simple beam bridge with a T-shaped cross-section. The total length and height of the bridge are 1260 mm and 140 mm, respectively. The top width of the bridge is 100 mm while the bottom width is 50 mm. Eleven diaphragms were placed inside the bridge cross-section every 105 mm to increase its strength. The top flange was also constructed with two layers to increase the bridge's strength. An image of our bridge can be seen in *Figure 1*.



Figure 1: The final bridge design

Key Design Decisions

Bridge Length: 1260 mm

- This is the shortest possible length the group was comfortable would fit on the testing apparatus. A smaller length would have been preferable to reduce the amount of material used.

Bridge Height: 140 mm

- This is the tallest possible height that was selected and that remained within the material constraints. A taller height would have been more preferable to maximize the second moment of area of the bridge's cross-section.

Track Width: 100 mm

- This is the smallest width allowed for the track as per the requirements. This could not have been further optimized.

Track Thickness: 2.54 mm (2 layers of matboard)

- This is the thickest track possible that was selected and that remained within the material constraints. A thicker track would have been preferable to maximize the second moment of area of the bridge's cross-section, and to strengthen the track against compression.

Diaphragm Spacing: 105 mm

- This is the greatest number of diaphragms possible that were selected and that remained within the material constraints and the cross-section and length selected. More diaphragms would have been preferable to strengthen the webs of the bridge against local buckling.

Splice Location: $\sim\frac{1}{4}$ of the way down the bridge (280 mm)

- This is the optimal location for a splice, because it minimizes the shear force and bending moment at the splice location. Minimizing the shear force and bending moment at the splice is important because the splice is the weakest point of the beam [1].

Iterations

Iteration 0: Design 0

This is the base design provided in the instructions document for this project. The Design 0 bridge is 1200 mm long and has four diaphragms spaced 400 mm apart. Its cross-section and safety factors are shown in *Figure 2* and *Figure 3*.

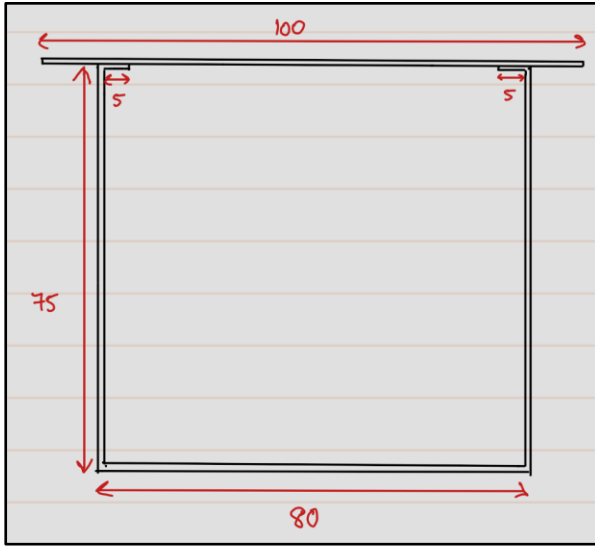


Figure 2: Design 0 cross-section with dimensions in millimeters. Thickness of matboard in the diagram is 1.27 mm (too small to label in the drawing).

FOS:	
Max. Tension:	3.878749
Max. Compression:	0.922537
Max. Shear Force:	2.259602
Shear at Glue Tabs:	7.952565
Case 1:	0.554256
Case 2:	3.768941
Case 3:	4.721797
Case 4:	2.902297
Min. FOS:	Case 1
Min. FOS Value:	0.554256
Max. Train Weight:	250.523751 N

Figure 3: MATLAB code output for Design 0. The design is weakest against compression and case one local buckling (buckling of the compressive flange between the webs).

NOTE: For the following iterations, the length of the bridge was changed to 1260 mm, and the total height of the bridge was changed to 80 mm. These changes were not considered as their own iterations since they were minor modifications made to meet the constraints of the assignment.

Iteration 1: Add a second layer of matboard to the top flange

GOAL: Strengthen the bridge against compression and case one local buckling

To strengthen the bridge against compression and case one local buckling, a second layer of matboard was added to the top flange as shown below in Figure 3. This was done to increase the second moment of area, I , for the cross-section, since it concentrates the bridge's mass further from the centroidal axis, y_{bar} . This modification also moves the location of the centroidal axis upwards, since the mass is specifically increased at the top of the cross-section. In the equation, $\sigma = \frac{My_{top}}{I}$, for compressive stress, the distance between the top flange and the centroidal axis, y_{top} , decreases, while, I , the second moment of area slightly increases (due to the additional thickness of the top layer). As a result, a much larger load is required to reach failure as shown in Figure 4. Thickening the top flange also increases the bridge's strength against case 1 local buckling, because the case 1 flexural buckling capacity is proportional to the square of the thickness of the top flange.

Overall, this increases the bridge's strength against compression and case one buckling. The bridge now fails due to compression when its stress surpasses six megapascals, or the maximum compressive strength of the matboard.

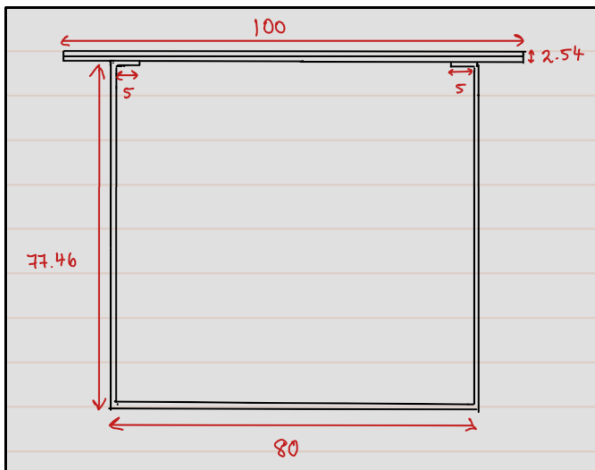


Figure 3: Iteration 1 cross-section with dimension in millimeters.

FOS:	
Max. Tension:	4.029972
Max. Compression:	1.417804
Max. Shear Force:	2.301613
Shear at Glue Tabs:	6.663207
Case 1:	3.407243
Case 2:	23.169249
Case 3:	10.638095
Case 4:	2.696459
Min. FOS:	Compression
Min. FOS Value:	1.417804
Max. Train Weight:	640.847391 N

Figure 4: MATLAB code output for Iteration 1. Thickening the top layer has improved strength against case one buckling and compression.

Iteration 2: Increase the total height of the bridge to 140 mm by increasing the web height by 60 mm

GOAL: Strengthen the bridge against compression

The next modification that was made was to increase the web height to 60 mm, which increased the total bridge height to 140 mm as shown in *Figure 5*. This was done because increasing the height of the webs increases the second moment of area, I , for the cross-section, since it is roughly proportional to the cube of the height of the webs. This strengthens the bridge against compression for the same reason discussed in *Iteration 1* and can be seen in *Figure 6*.

After applying the modification, the lowest factor of safety (FOS) was now 1.61 as shown in *Figure 6* and was caused by case four buckling, or the shear buckling of the webs.

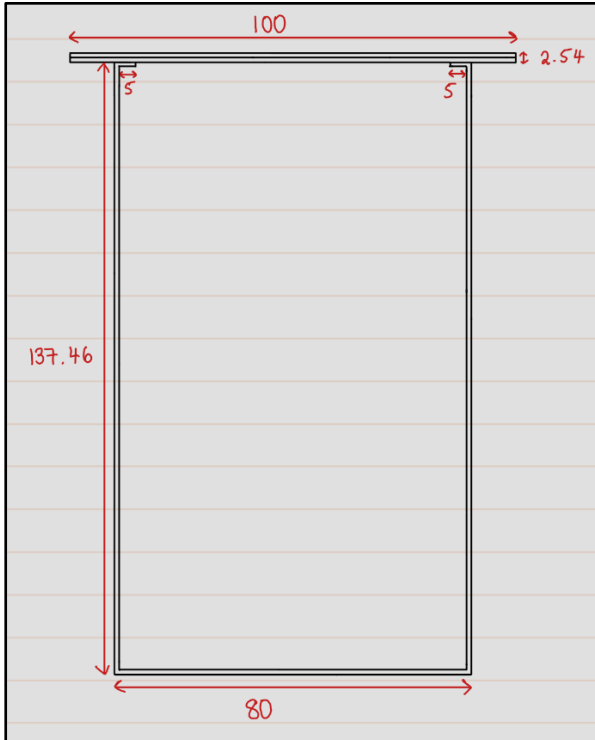


Figure 5: Iteration 2 cross-section with dimensions in millimeters.

FOS:	
Max. Tension:	8.727207
Max. Compression:	2.734636
Max. Shear Force:	3.914631
Shear at Glue Tabs:	12.581921
Case 1:	6.571831
Case 2:	44.688454
Case 3:	5.557320
Case 4:	1.616327
Min. FOS:	Case 4
Min. FOS Value:	1.616327
Max. Train Weight:	730.579760 N

Figure 6: MATLAB code output for Iteration 2. Increasing the height of the webs has strengthened the bridge against compression, and the lowest FOS is now case four buckling.

Iteration 3: Increase the width of each glue tab to 10 mm

GOAL: Strengthen the glue tabs against shear buckling of the webs

The next modification that was made was to increase the width of the glue tabs as shown in Figure 7. This was done to strengthen them against shear. This was done because the glue tabs fail due to shear when the shear stress reaches two megapascals, or $\frac{VQ}{Ib} = 2 \text{ MPa}$. Therefore, by increasing the width of the glue tab, b , a greater load, V , is required to reach failure. As seen in the MATLAB code output in Figure 8, widening the glue tabs strengthened them against shear as expected; however, this change also slightly decreased the predicted failure load of the bridge compared to the previous iteration. Nonetheless, this modification was kept to account for the possibility that the glue might not achieve full strength (2MPa) at the time of the bridge testing.

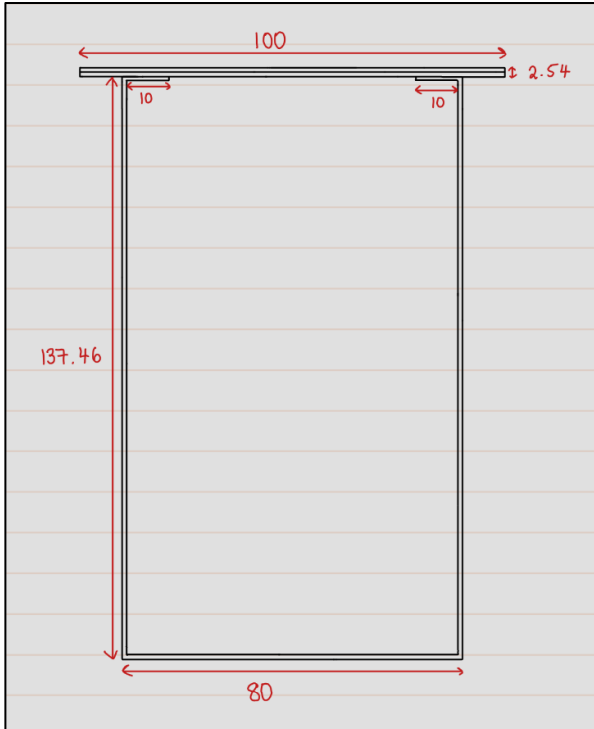


Figure 7: Iteration 3 cross-section with dimensions in millimeters.

FOS:	
Max. Tension:	8.771303
Max. Compression:	2.823780
Max. Shear Force:	3.912636
Shear at Glue Tabs:	23.361886
Case 1:	6.786061
Case 2:	46.145215
Case 3:	5.936819
Case 4:	1.615503
Min. FOS:	Case 4
Min. FOS Value:	1.615503
Max. Train Weight:	730.207454 N

Figure 8: MATLAB code output for Iteration 3. Increasing the width of the glue tabs has caused the FOS for shear at glue tabs to increase.

Iteration 4: Decrease the distance between diaphragms to 126 mm

GOAL: Strengthen the bridge against case four shear buckling of the webs

At this point in the design process, the diaphragm spacing had still not been modified from Design 0. To add more diaphragms, the spacing of each diaphragm was reduced to 126 mm.

This value was chosen specifically to align evenly with the total length of the bridge as $\frac{1260 \text{ mm}}{126 \text{ mm}} = 10$ evenly spaced bridge sections split by diaphragms. A smaller spacing was chosen to increase the FOS for case four buckling, which was currently the bridge's weakest point. The case four shear buckling is inversely proportional to the square of the distance between the diaphragms, so when this distance is smaller, the flexural buckling capacity is higher. This drastically improved the case four shear buckling capacity as shown in Figure 9 and the new lowest FOS was compression.

```

FOS:
    Max. Tension:            8.771303
    Max. Compression:        2.823780
    Max. Shear Force:        3.912636
    Shear at Glue Tabs:      23.361886
    Case 1:                  6.786061
    Case 2:                  46.145215
    Case 3:                  5.936819
    Case 4:                  3.215993

Min. FOS:                    Compression
Min. FOS Value:              2.823780
Max. Train Weight:          1276.348670 N

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Figure 9: MATLAB code output after the distance between diaphragms was decreased to 126 mm. The FOS for case four shear buckling has approximately doubled, and the FOS is now compression, resulting in a failure load of 1276 N.

Iteration 5: Decrease the width of the bottom of the cross-section to 50 mm

GOAL: Account for a construction error

Due to an error made during construction, the pieces intended for the main cross-section were measured and cut 30 mm shorter than the intended width of 80 mm. Decreasing the entire width of the bottom of the cross-section to 50 mm was the only way to incorporate this error while keeping the height of the bridge a multiple of 20 mm, maintaining the size of glue tab, and preventing the loss of material. The modified cross section can be seen in *Figure 10*.

While unfortunate and unintentional, this change affected the FOS's for all the failure modes to varying extents as shown in *Figure 11*. All the FOS's decreased, except for case 1 and case 3 local buckling, which increased slightly. Most changes to FOS's did not significantly impact the predicted performance of the bridge, and do not merit discussion or analysis. However, one significant change must be noted is that the FOS for compression decreased. Since this was already the lowest FOS, it caused the predicted failure load of our bridge to decrease from 1276 N to 1207 N as shown in *Figure 11*. This is because decreasing the width of the cross-section caused the second moment of area, I , to decrease. The bridge would now fail due to compression when its stress surpasses six megapascals. Therefore, a smaller load, or a smaller bending moment is required for the bridge to reach failure.

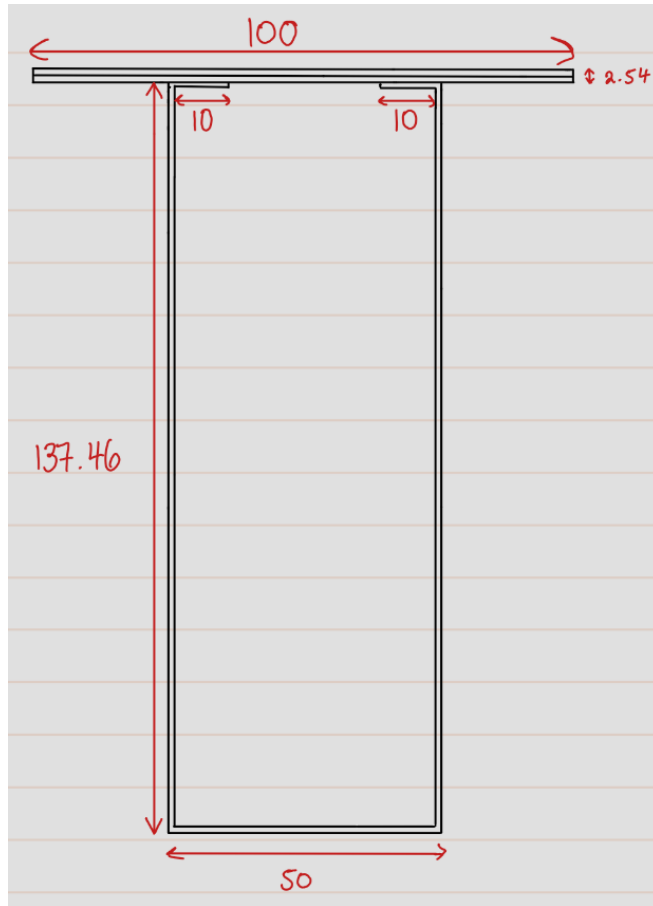


Figure 10: Iteration 5 cross-section with dimensions in millimeters.

Iteration 6: Decrease the distance between diaphragms to 105 mm

GOAL: Strengthen the bridge against case four shear buckling of the webs

During construction, it became clear that once decreasing the bottom width to 50 mm, there was more material to spare. The decision to add more diaphragms to the bridge using the additional material was made, as it seemed to make a drastic difference in improving the case four shear buckling of the webs in iteration four. Unfortunately, adding these diaphragms did not impact the failure load, since the predicted failure mode at this point was compression, which would not improve with the diaphragms. However, implementing the extra diaphragms did increase the FOS for case four shear buckling for the same reason as discussed in Iteration 4. While it did not help with the compression, this modification was still somewhat favourable as the case four shear buckling was our second-lowest FOS value.

FOS:

Max. Tension:	7.168335
Max. Compression:	2.670448
Max. Shear Force:	3.776395
Shear at Glue Tabs:	22.145383
Case 1:	16.428992
Case 2:	6.982322
Case 3:	6.787699
Case 4:	3.104010

Min. FOS:	Compression
Min. FOS Value:	2.670448
Max. Train Weight:	1207.042403 N

Figure 11: MATLAB code output after the width of the bottom section was decreased to 50 mm. This change caused the compression FOS and failure load to decrease.


```

FOS:
    Max. Tension:                7.168335
    Max. Compression:            2.670448
    Max. Shear Force:            3.776395
    Shear at Glue Tabs:          22.145383
    Case 1:                      16.428992
    Case 2:                      6.982322
    Case 3:                      6.787699
    Case 4:                      3.858576

Min. FOS:                        Compression
Min. FOS Value:                  2.670448
Max. Train Weight:               1207.042403 N

```

Figure 12: MATLAB code output after the number of diaphragms was increased. Maximum train weight and minimum FOS are the same, but the case 4 shear buckling FOS increased by ~ 0.9 .

NOTE: There is a discrepancy between the calculations, the design decision for *Iteration 6*, and the engineering drawings. While it was intended that the diaphragms be placed at a spacing of 105 mm in *Iteration 6* and is also what exists in the final bridge design shown in the engineering drawings document, after constructing the bridge, it was realized that the diaphragms were designed to only attach at the top and bottom of the cross section, undermining their entire purpose. The diaphragms were intended to reduce the case four local shear buckling. However, to fulfill their purpose, they had to be attached to the sides of the cross-section with some kind of glue tab. Unfortunately, this was a key step that was missed and that defeated their main purpose.

While the purpose of the glue tabs was to strengthen the bridge, in reality, the bridge is still extremely weak against local shear buckling. As a result, the predicted load it can carry is drastically decreased which can be seen in the calculations document.

In the future, side glue tabs should be implemented on the diaphragms to improve the bridge design and ensure each component of the bridge performs its intended function.

References

- [1] "What is Beam Splice? Why, Where, When & How," *SteelPRO PEB*, Feb. 19, 2025.
<https://peb.steelprogroup.com/steel-structure/components/beam-splice/>